

Report of Project 1: Thermochemistry

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1 Specific Heats

1.1 Normalized Constant-Volume and Pressure Specific Heat for Diatomic Molecules - Q1(d)

The constant-pressure and constant-volume specific heats are calculated from CHEMKIN for the species H_2 , N_2 , O_2 , and CO ; the data is then normalized by R (specific gas constant). Fig. 1 shows the plot of the $\frac{c_p(T)}{R}$ and $\frac{c_v(T)}{R}$ versus T for the species H_2 , N_2 , O_2 , and CO .

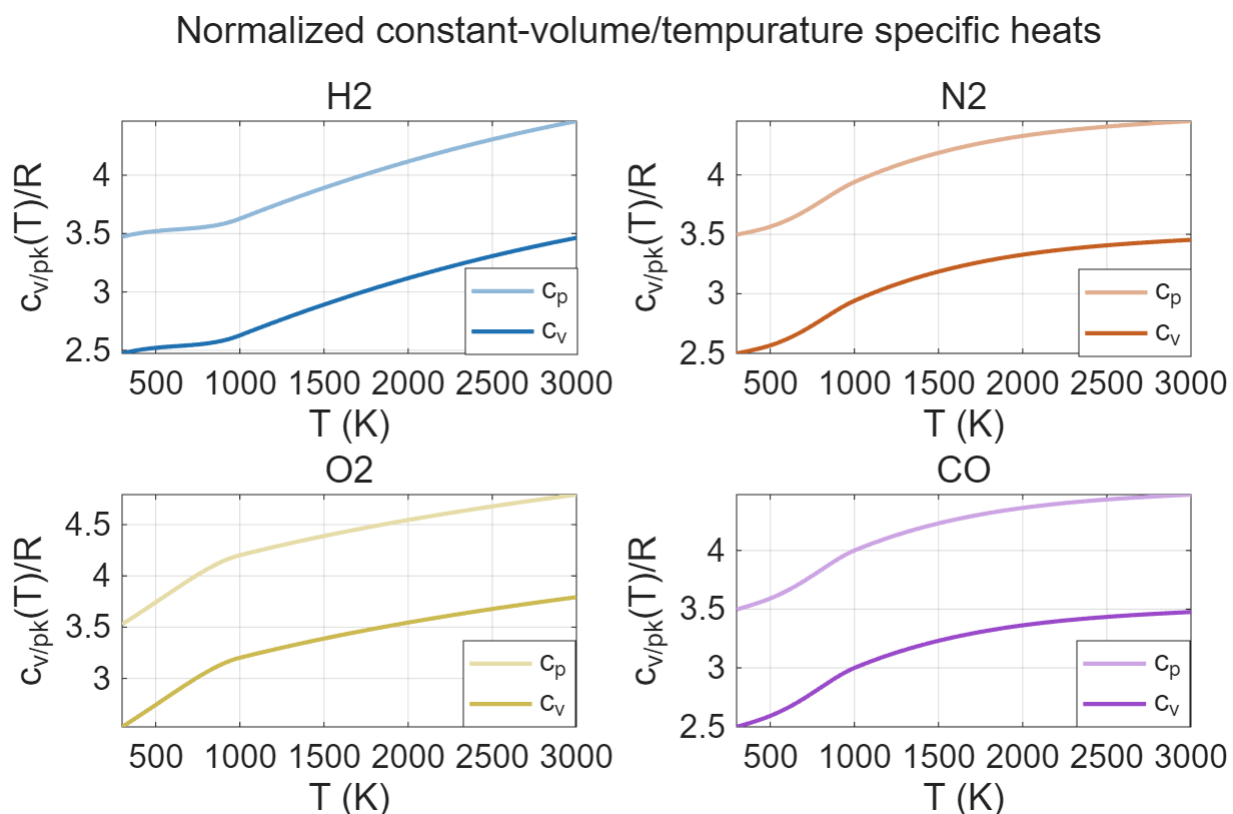


Figure 1 Normalized Constant Volume and Temperature Specific Heats for Each Species

The values for $c_p(T)$ and $c_v(T)$ for each species is offset by exactly 1 normalized unit of $\frac{c_{p/vk}(T)}{R}$ with $c_p(T)$ being the larger value of $1 + c_v(T)$. This relationship stays constant throughout the range of temperatures.

$$c_p(T) - c_v(T) = R \quad (1)$$

Eq. (1) shows the relationship between $c_p(T)$ and $c_v(T)$ for ideal gases. If this gets normalized by dividing by R and rearranging the equation, you get Eq. (2).

$$\frac{c_p(T)}{R} = 1 + \frac{c_v(T)}{R} \quad (2)$$

Eq. (2) is consistent with Fig. 1, and explains this relation shown in the figure.

1.2 Specific Heat Ratio for Diatomic Molecules – Q1(e)

The specific heat ratio can be modeled by Eq. (3) below.

$$\gamma = \frac{c_p(T)}{c_v(T)} \quad (3)$$

Fig. 2 shows this information for the species H_2 , N_2 , O_2 , and CO , and the results are consistent as linear results are to be expected. An important note is that each molecule has overlying data, so some of the data is not visible.

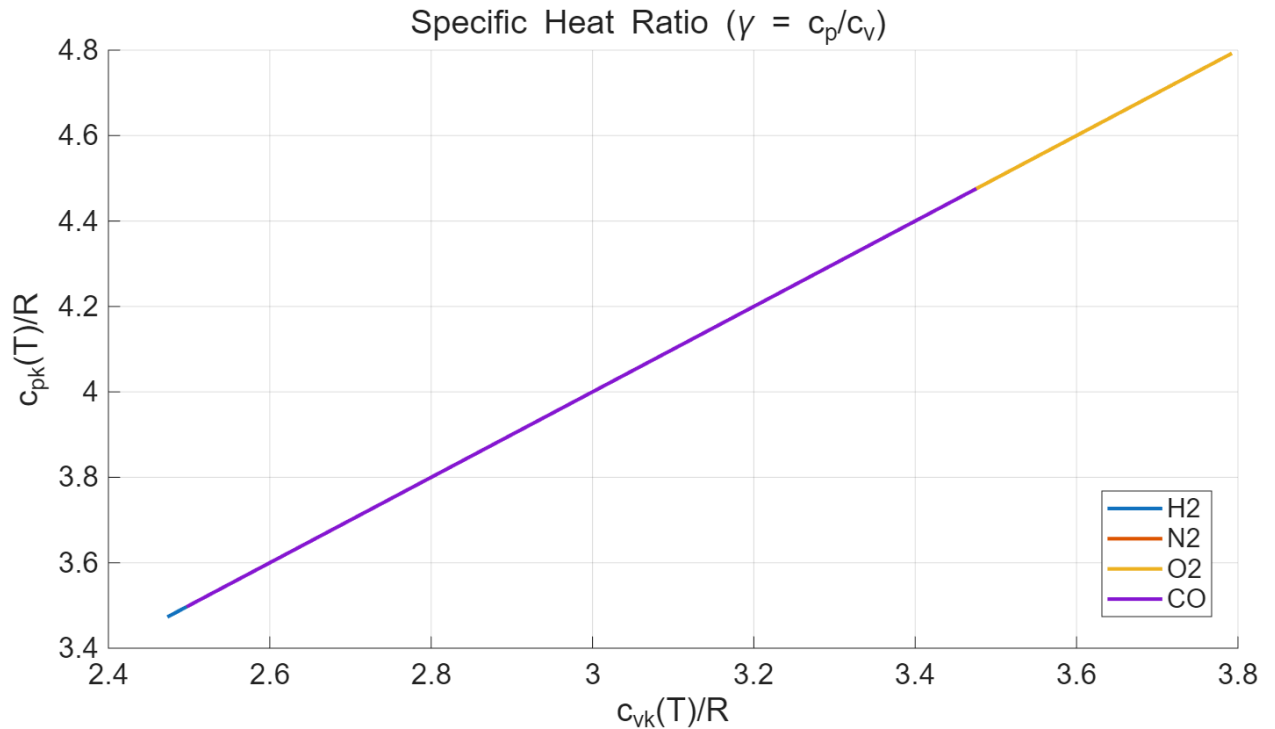


Figure 2 Specific Heat Ratio for H_2 , N_2 , O_2 , and CO

The specific heat ratio does change as temperature changes. This relation is shown in Fig. 3 when the specific heat ratio is plotted against the temperature.

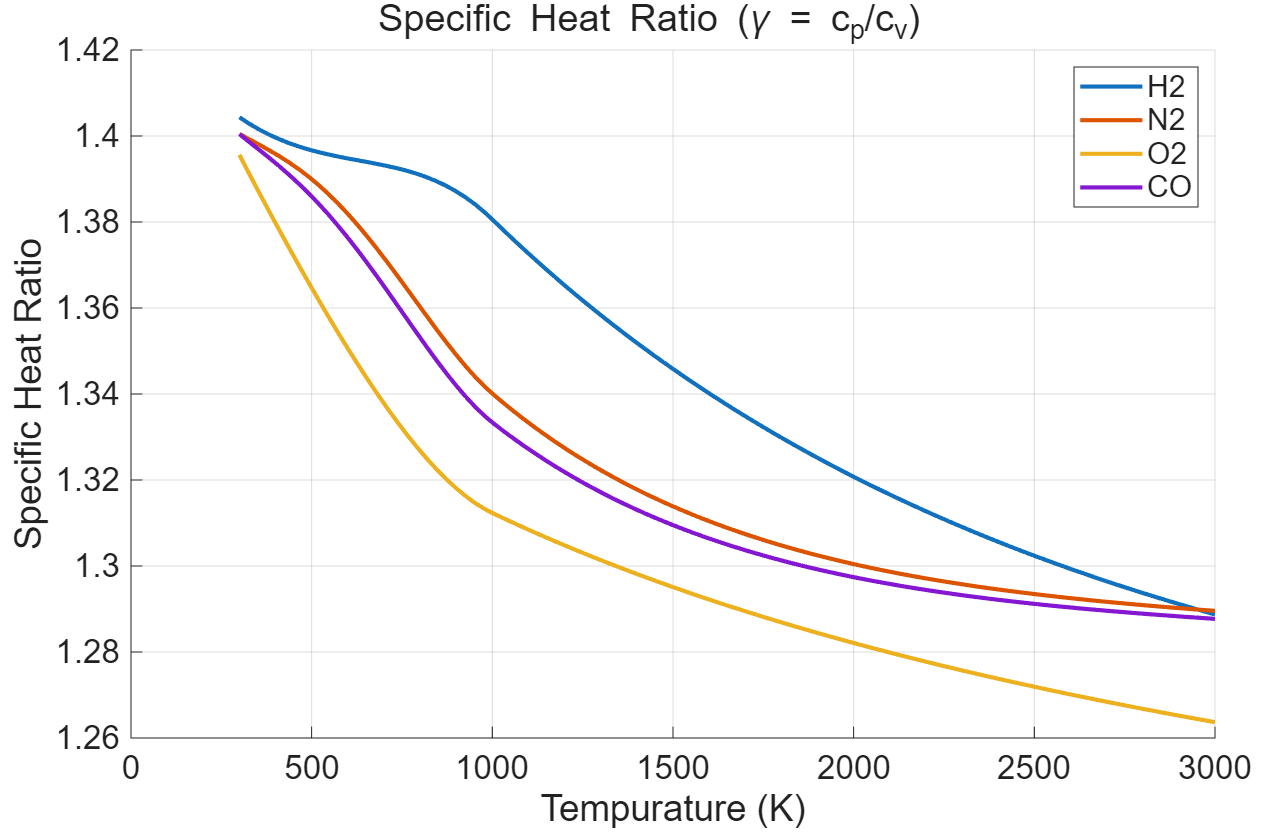


Figure 3 Specific Heat Ratio versus Temperature for H₂, N₂, O₂, and CO

Under temperatures of 500K, the specific heat ratio is around 1.4. This result is normal, as ideal diatomic gases have a specific heat ratio of 1.4 for regular working temperatures. As the temperature increases, the specific heat ratio decreases to values between 1.28 and 1.3 which is also to be expected as the gases become higher energy due to an increase in thermal energy, so c_v will increase faster than c_p does.

1.3 Mathematical Model of $\frac{c_v(T)}{R}$ for Diatomic Molecules - Q1(f)

Eq. (4) is a model for diatomic models that calculates $\frac{c_v(T)}{R}$ where $\tau = \frac{\theta_v}{T}$ and θ_v is the vibrational temperature.

$$\frac{c_v(T)}{R} = \frac{5}{2} + \frac{\tau^2 e^{-\tau}}{(1 - e^{-\tau})^2} \quad (4)$$

The diatomic molecules H₂, N₂, O₂, and CO are used in this equation with vibrational temperatures of 6332K, 3374K, 2256K and 3103K, respectively. Fig. 4 shows this model with the CHEMKIN $\frac{c_v(T)}{R}$ values obtained with respect to temperature.

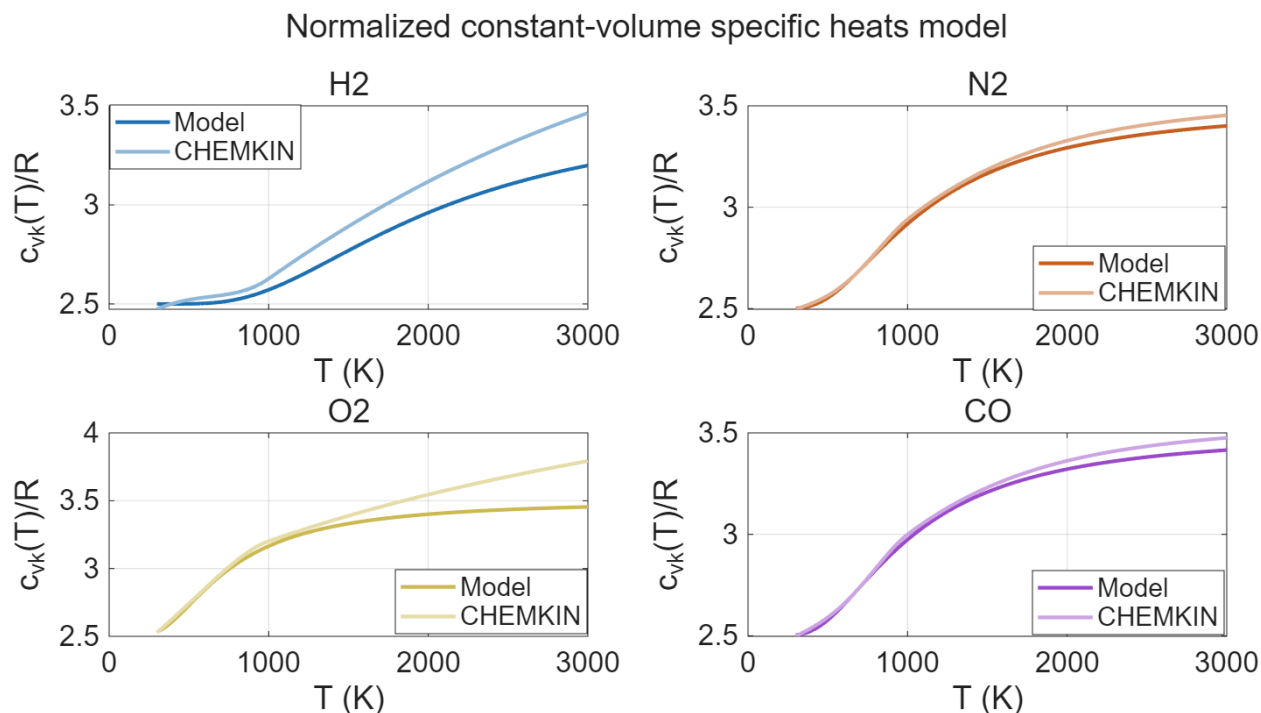


Figure 4 Normalized Constant-Volume specific heats with model for diatomic molecules

The model is consistent with the CHEMKIN results with varying accuracy. Table 1 shows this relation with the percent difference between each species at varying temperatures. CHEMKIN also consistently records higher $c_v(T)$ values than the model.

Table 1 Percent Difference for CHEMKIN and Given Model at Different Temperatures for Each Species

Temperature	H ₂	N ₂	O ₂	CO
300K	1.0818	0.1861	0.1183	0.2458
900K	1.5434	0.6114	1.1693	0.7182
1500K	4.3163	0.6052	1.7189	0.7009
2100K	5.3790	1.1462	4.6967	1.3126
2700K	7.0147	1.4415	7.8559	1.6427
3000K	7.9455	1.5336	9.3373	1.7524

Roughly speaking, the model works well with temperatures less than 1500K for all species with percent errors around 1% except H₂. The model does not work well for H₂ as it is exceedingly small with a low molecular weight. This makes it behave slightly differently than other diatomic molecules such as N₂, O₂, and CO. At higher temperatures, the model does not predict the trends that well. This is to be expected as gases become higher energy due to an increase in thermal energy, so c_v will increase more than expected which explains why the model consistently records lower values at higher temperatures.

1.4 Normalized Constant-Volume and Pressure Specific Heat for Triatomic Molecules - Q1(g)

The constant-pressure and constant-volume specific heats are calculated from CHEMKIN for the species H_2O , HO_2 , CH_2 , HCO , and CO_2 ; the data is then normalized by R (specific gas constant).

Fig. 5 shows the plot of the $\frac{c_p(T)}{R}$ and $\frac{c_v(T)}{R}$ versus T for the species H_2O , HO_2 , CH_2 , HCO , and CO_2 .

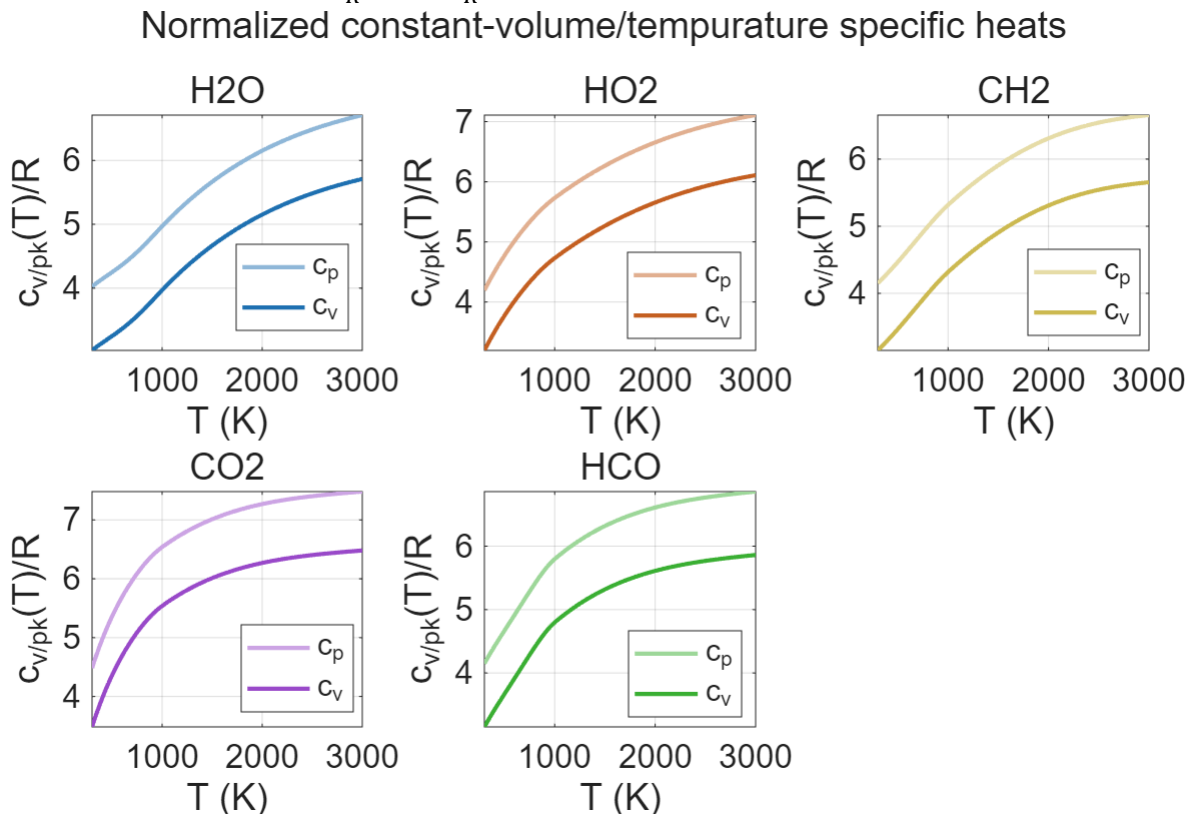


Figure 5 Normalized Constant Volume and Temperature Specific Heats for Triatomic Molecules

The results are consistent with the results for the diatomic molecules (section 1.1). This is where the values for $c_p(T)$ and $c_v(T)$ for each species is offset by exactly 1 normalized unit of $\frac{c_{p/vk}(T)}{R}$ with $c_p(T)$ being the larger value of $1 + c_v(T)$.

$$\frac{c_p(T)}{R} = 1 + \frac{c_v(T)}{R} \quad (5)$$

This relationship stays constant throughout the range of temperatures with Eq. (5). showing the relationship between the specific volume and specific temperature.

2 Chemical Equilibrium: Hydrogen-Air

2.1 Minor Species with Significant Mass Fractions: Chemical Equilibrium of N_2 and all relevant O-H Species – Q2(1)

The major species are H_2 , N_2 , O_2 , and H_2O as these molecules are present in higher amounts, and the minor species are H , O , OH , HO_2 , and H_2O_2 . The mass fraction is considered significant if the mass fraction is greater than 10^{-6} . According to Table 2, which shows the maximum mass fraction of each minor species, it is evident that all the relevant O-H species except for H_2O_2 are considered significant.

Table 2 Maximum Mass Fractions of Different Minor Species

Species	H	OH	O	HO_2	H_2O_2
Maximum Y_k	1.5415e-4	4.8981e-3	3.9779e-4	3.3474e-06	2.4287e-07

These results can also be plotted in a semi-log plot with mass fraction of each element on the y-axis and the mixture fraction on the x-axis. The semi-log plot is used as it is easier to see slight changes in value compared to the linear model. Fig. 6 shows this plot for all minor O-H species.

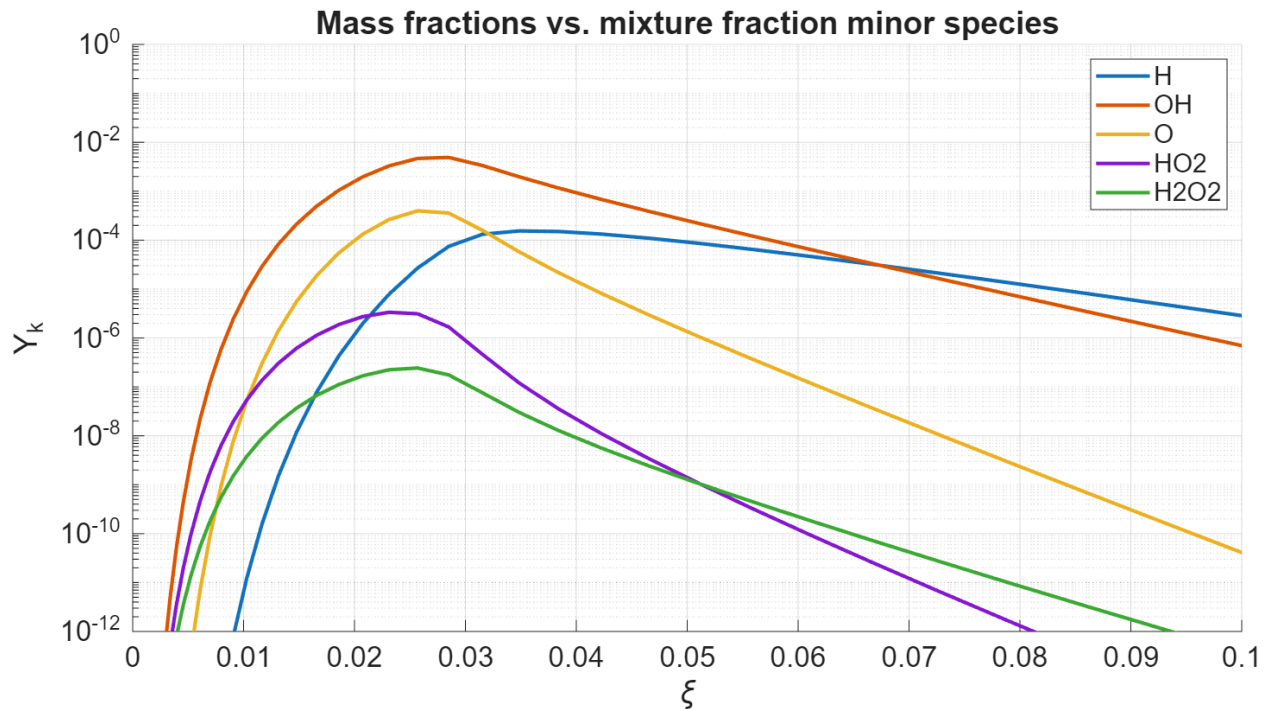


Figure 6 Mass Fractions of all Relevant Minor O-H Species

2.2 Chemical Equilibrium of all Relevant O–H and N–O Species: Relative Magnitudes of the Oxides of Nitrogen, NO, N₂O, NO₂ – Q2(2)

One of the ways to see the relative magnitudes of the oxides in nitrogen is to look at the semi-log graph of the mass fractions vs mixture fraction of each. Fig. 7 below shows this, and the semi-log scale is used to make the relationship clearer as the values are small.

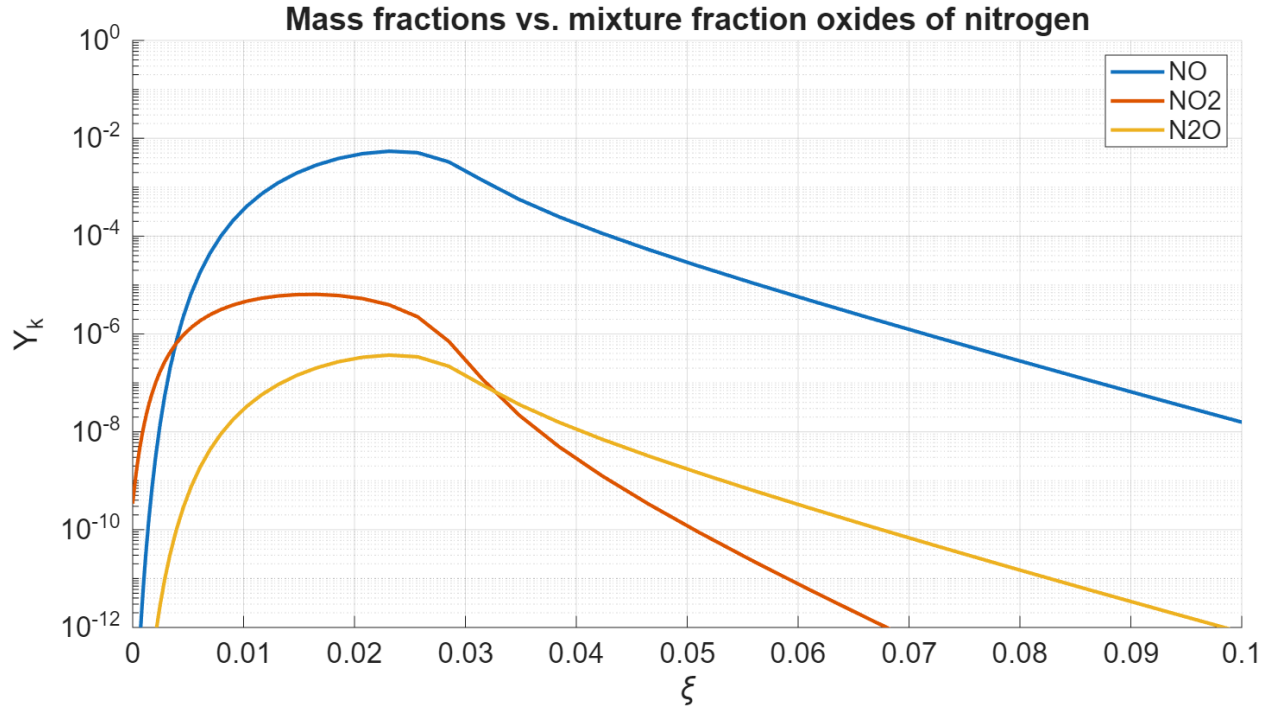


Figure 7 Mass Fractions vs Mixture Fractions Semi-log Plot of NO, N₂O, NO₂

The maximum mass fraction of NO is 0.0054345 at a mixing fraction of 0.023112. For NO₂, the maximum mass fraction is 6.436e-6 at a mixing fraction of 0.016595, and for N₂O, the maximum mass fraction is 3.6938e-7 at a mixing fraction of 0.023112. NO is present in considerable amounts than NO₂ which is present in higher amounts than N₂O for fuel lean states. N₂O and NO₂ switch locations as the mixture gets more fuel rich. These maximums all occur at fuel-lean states. One reason for this is due to the abundance of oxygen present in more fuel-lean mixtures which causes more nitrogen oxides.

2.3 Chemical Equilibrium of all Relevant O–H–N Species– Q2(3)

Model E, the chemical equilibrium of all relevant O–H–N species, has several interesting features.

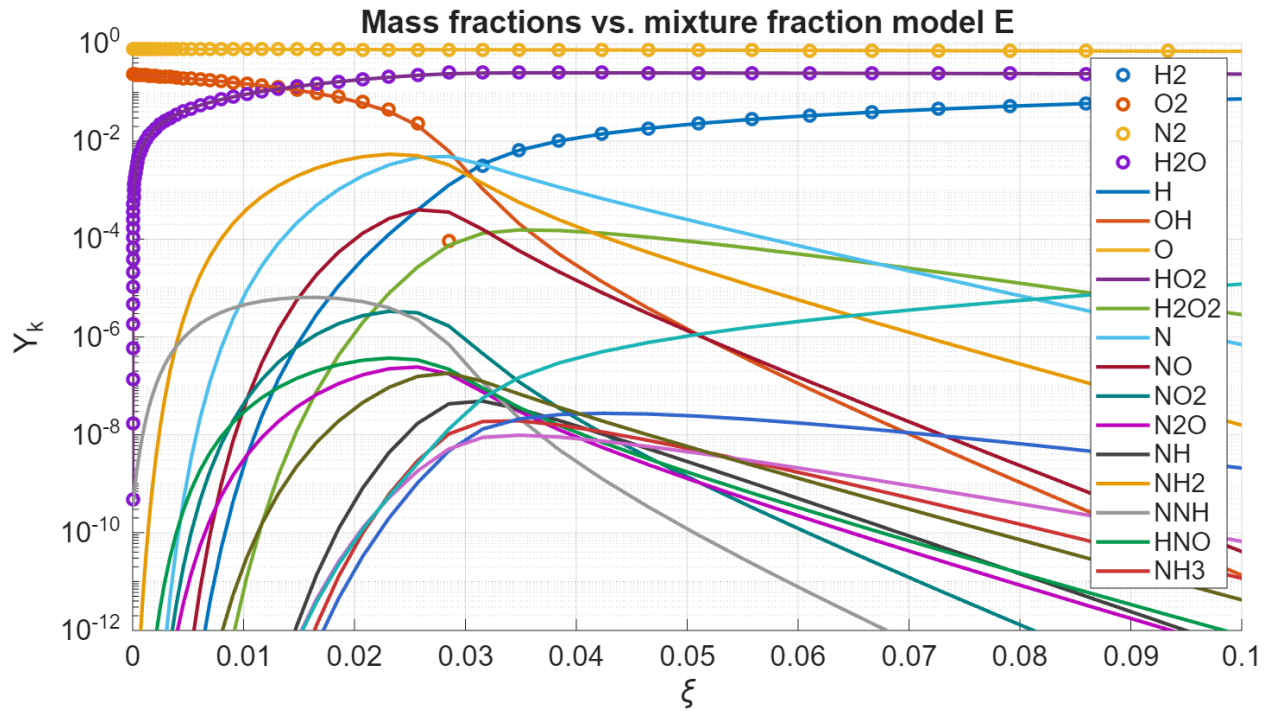


Figure 8 Mass Fractions vs Mixture Fraction of All Relevant O–H–N Species

Fig. 8 shows the mass fraction vs mixture fraction for each major and minor species present. The major species, H_2 , N_2 , O_2 , and H_2O are present in exceptionally substantial amounts with magnitudes greater than $10e-3$. The next group of species that has the next greatest mass fraction is all the relevant O–H species (H , OH , O , HO_2 , H_2O_2) with maximum mass fractions greater than $10e-4$. The next group of species is the N–O species (N , NO , NO_2 , N_2O) with maximum mass fractions greater than $10e-7$. Lastly, the O–H–N (NH , NH_2 , NNH , HNO , NH_3) species are present in the least amount with maximum mass fractions greater than $10e-9$. The mixture fraction of the maximum mass fraction of each group also moves towards more fuel rich. This is due to the burning of oxygen as the mixture gets more fuel rich.

2.4 Variation of the Peak Temperature Given by the Different Models- Q2(4)

Each model has different peak temperatures. Table 3 below shows these variations.

Table 3 Maximum Temperature of Each Model

Model	complete combustion	chemical equilibrium of H_2 , O_2 , N_2 , H_2O	chemical equilibrium of N_2 and all relevant O–H species	chemical equilibrium of all relevant O–H and N–O species	chemical equilibrium of all relevant O–H–N species
Max Temperature Equilibrium	2525.1 K	2446.4 K	2399.8 K	2394 K	2394 K

The maximum equilibrium temperature follows an exponential decay trend as more types of species get added. This trend goes to 2394 K with the last two species groups to be added as shown in Table 3. This trend is due to each increasing model taking more energy to complete the reaction which pulls away from the energy being transferred into heat and thus temperature.

3 Chemical Equilibrium: Methane-Air

3.1 Complete-Combustion Model of the Major Species for Rich and Lean Mixtures – Q3(1)

The mass fraction of the major species H_2 , N_2 , O_2 , H_2O , CH_4 , CO_2 , and CO can be plotted over the mixture fraction. The dotted points are the complete-combustion model while the lines are the equilibrium composition.

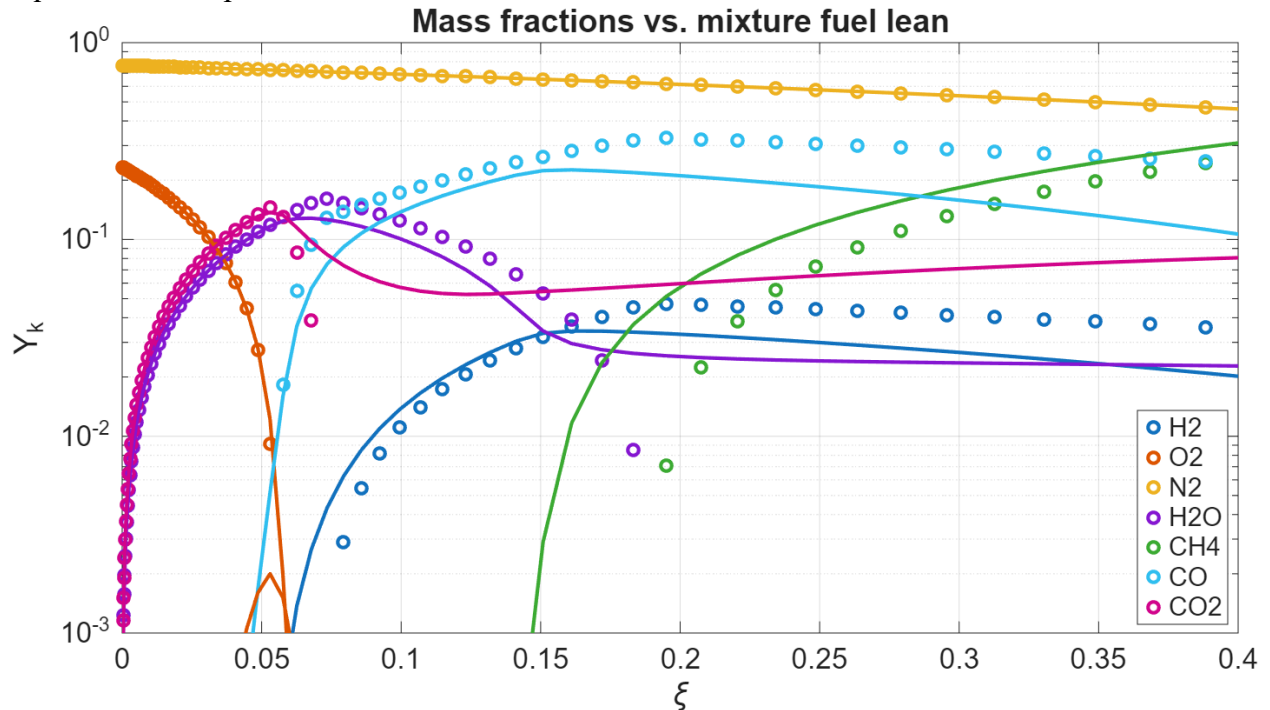


Figure 9 Mass Fractions vs Mixture for Major Species in Fuel Lean Conditions

Fig. 9 shows the mass fractions when the mixture is fuel lean. The complete-combustion model describes the equilibrium composition well. The species O_2 , N_2 , CO_2 , and H_2O are described well as the complete-combustion data does not stray far from the equilibrium composition. However, the other major species are not depicted as well. This is since there is an assumption of perfect efficiency in the complete-combustion model. This is a fairer assumption for O_2 , N_2 , CO_2 , and H_2O , as these are assumed to be products, but not for the other species.

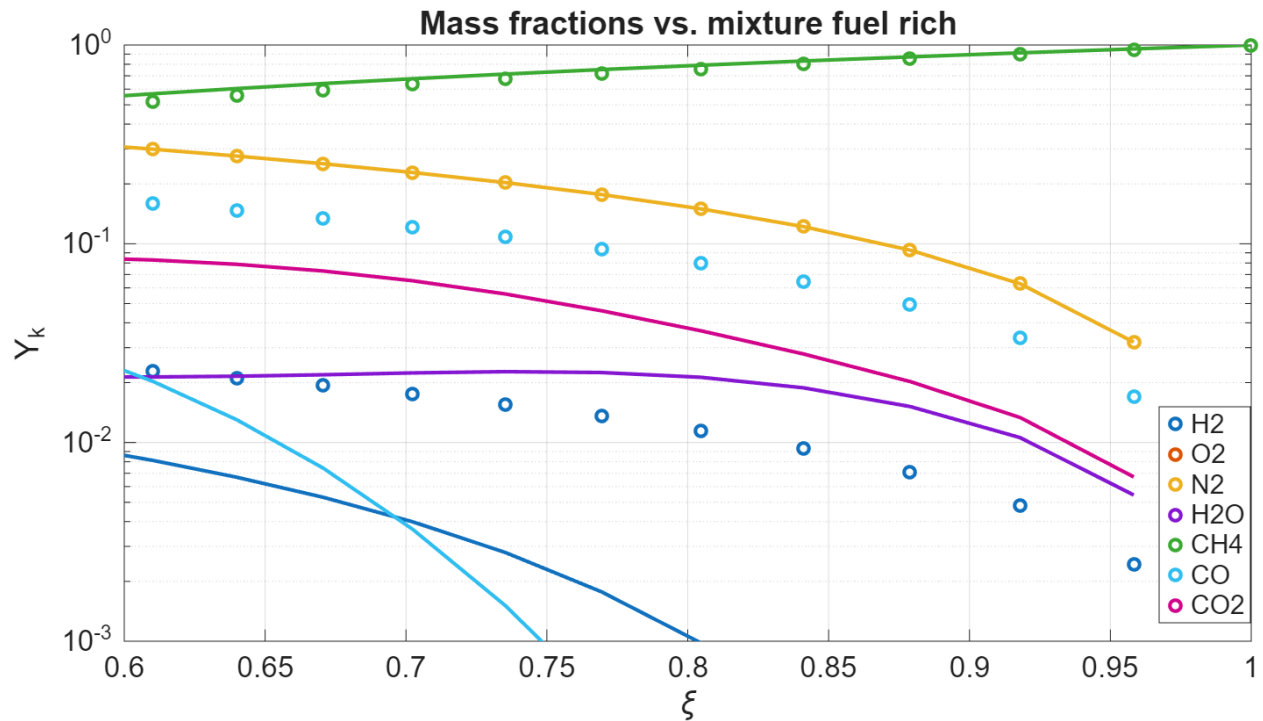


Figure 10 Mass Fractions vs Mixture for Major Species in Fuel Rich Conditions

Fig. 10 shows the mass fractions when the mixture is fuel rich. The complete-combustion model describes the equilibrium composition not as well as it did for fuel lean. The species N₂ and CH₄, are described well as the complete-combustion data does not stray far from the equilibrium composition. However, the other major species are not depicted as well. This is since there is an assumption of perfect efficiency in the complete-combustion model. However, in fuel-rich conditions, there is a lack of oxygen, noted by the fact that O₂ does not even appear as it is only present in negligible quantities. This lack of oxygen means the reaction cannot be fully efficient which leads to discrepancies in the models.

3.2 Comparison of Minor O-H Species, Hydrogen to Methane Flame - Q3(2)

The mass fractions of the minor O-H species, H, OH, O, HO₂, and H₂O₂, are different for methane-air and hydrogen-air reactions.

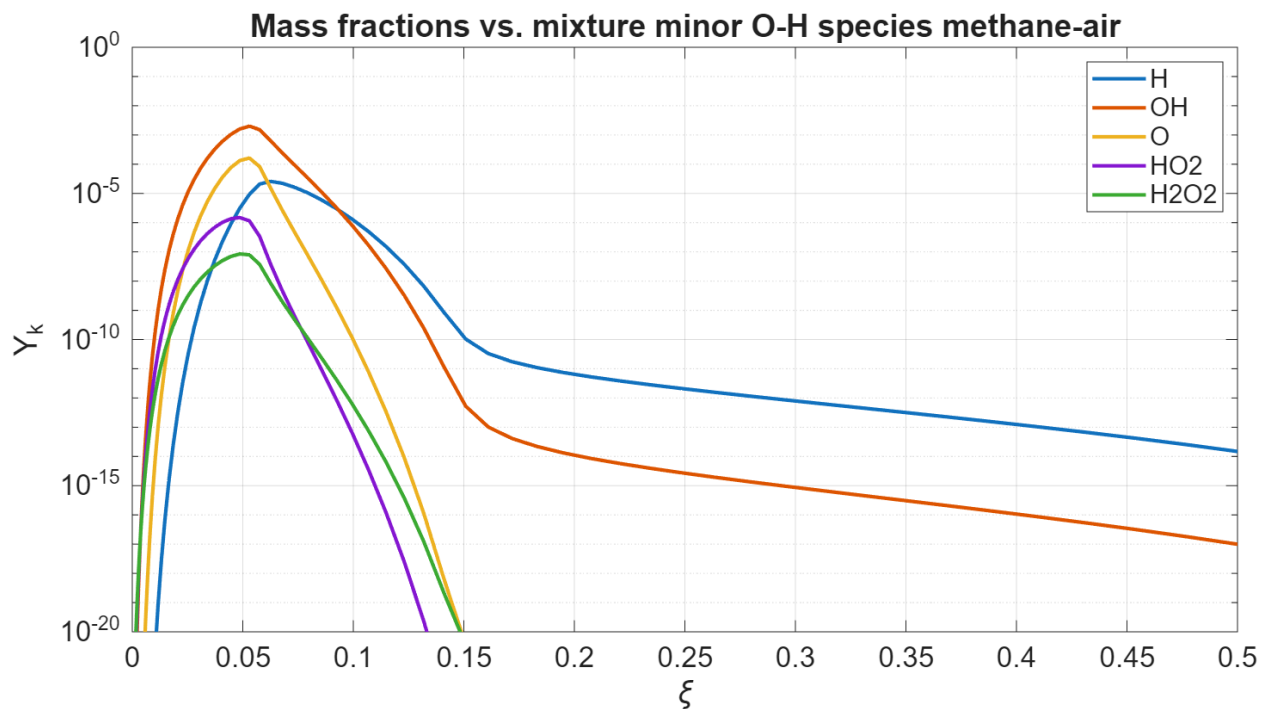


Figure 11 Mass Fractions vs Mixture Fraction in a Methane-Air Reaction of minor O-H Species

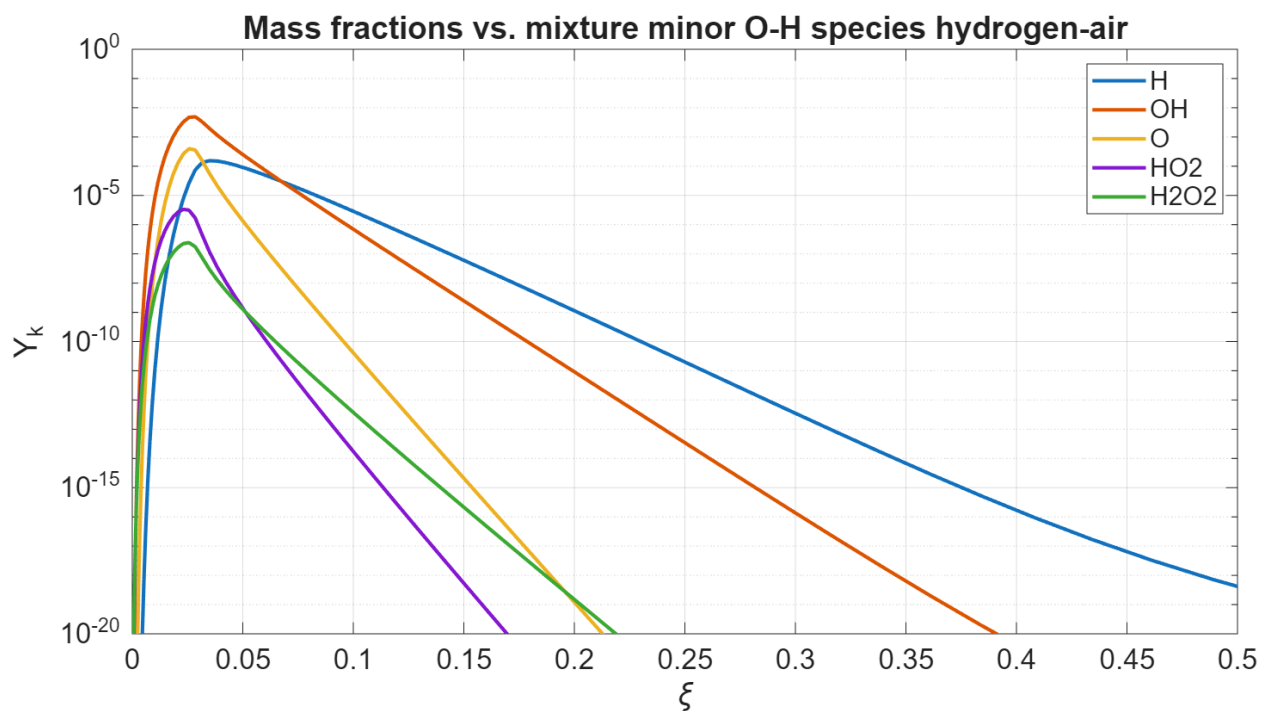


Figure 12 Mass Fractions vs Mixture Fraction in a Hydrogen-Air Reaction of minor O-H Species

Fig. 11 and Fig. 12 display these differences by showing the differences in mass fractions versus mixture fractions for each minor O-H species. These different mixtures have the same profile but with methane-air having a lower amount of the O-H species compared to the hydrogen-air mixture in more fuel rich conditions.

Table 4 Maximum Mass Fraction of Methane-Air and Hydrogen-Air Mixtures

Species	H	OH	O	HO₂	H₂O₂
Max Y_k methane-air	2.5811e-5	2.0008e-3	1.6274e-4	1.4983e-6	8.4775e-8
Max Y_k Hydrogen-air	1.5415e-4	4.8981e-3	3.5675e-4	3.347e-6	2.4287e-7

Table 4 shows the maximum mass fraction of each minor O-H species. The hydrogen-air mixture has consistently higher maximum mass fractions. For the methane-air mixture, the maximum mass fraction for each minor O-H species occurs at a mixture fraction of an average of 0.053. For hydrogen-air, this mixture fraction is on average 0.02813. This is interesting as even though methane-air has the maximum at a higher mixture fraction, there is a faster drop off as mixture fraction increases.

3.3 C₁-Species with Largest Mass Fraction – Q3(3)

Mass fraction vs mixture fraction for C₁ - species is displayed in Fig. 13 with significant importance on the 6 species with the largest mass fractions.

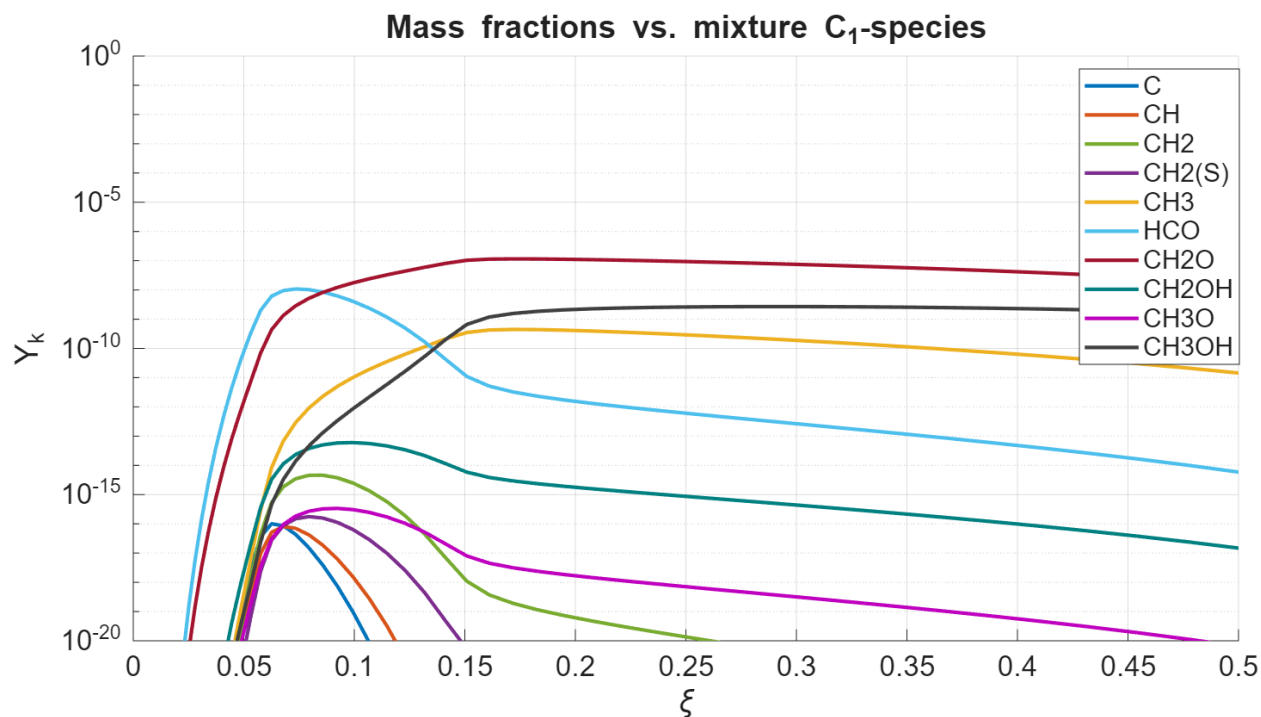


Figure 13 Mass Fraction vs Mixture Fraction for C_1 - Species

The figure shows the 6 C_1 - species with the largest mass fraction are CH_2O , HCO , CH_3OH , CH_3 , CH_2OH , and CH_2 . Table 5 narrows down on this information by showing the maximum mass fractions of each of the largest species.

Table 5 Maximum Mass Fractions of C_1 - Species Present in the Greatest Amount

Species	CH_2O	HCO	CH_3OH	CH_3	CH_2OH	CH_2
Maximum Y_k	1.1546e-7	1.0818e-8	2.6921e-9	4.4553e-10	6.0234e-14	4.6126e-15

All the other C_1 - species are present in amounts that are less than 10^{-15} as shown by Fig. 13.

3.4 C_2 -Species with Largest Mass Fraction – Q3(4)

Mass fraction vs mixture fraction for C_2 - species is displayed in Fig. 14 with significant importance on the 2 species with the largest mass fractions.

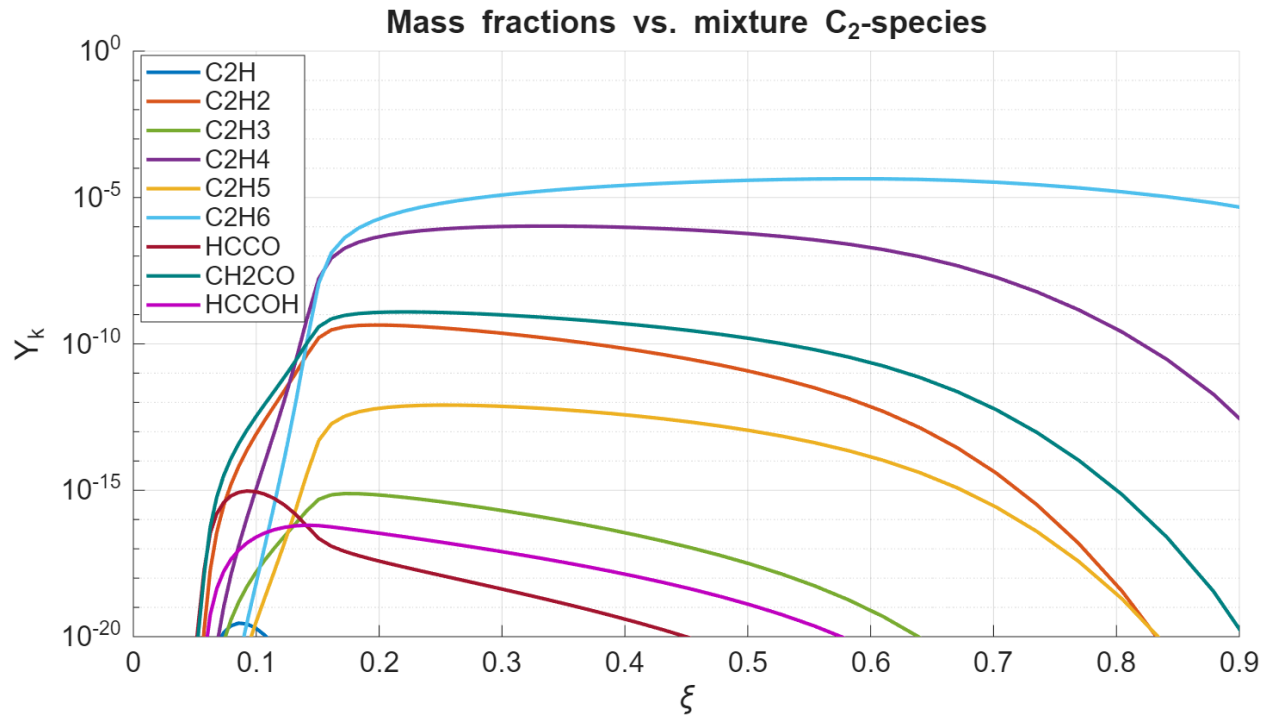


Figure 14 Mass Fraction vs Mixture Fraction for C_2 - Species

The figure shows the 2 C_2 - species with the largest mass fraction are C_2H_6 and C_2H_4 . Table 6 narrows down on this information by showing the maximum mass fractions of each of the largest species.

Table 6 Maximum Mass Fractions of C_2 - Species Present in the Greatest Amount

Species	C_2H_6	C_2H_4
Maximum Y_k	4.1207e-5	1.052e-6

All the other C_2 - species are present in amounts that are less than $10e-8$ as shown by Fig.14.